ITA06 - Machine Learning

EXP 1: Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis from a given set of training data samples.

Algorithm

1. Initialize hypothesis **h** with the most specific values (Ø).
2. Consider only positive training examples.
3. For each positive example:
    - If attribute matches hypothesis, keep it unchanged.
    - If attribute differs, replace it with **?**.
4. Ignore all negative examples.
5. Output the final hypothesis.

PROGRAM

```
# Training data

data = [

  ['Sunny','Warm','Normal','Strong','Warm','Same','Yes'],

  ['Sunny','Warm','High','Strong','Warm','Same','Yes'],

  ['Rainy','Cold','High','Strong','Warm','Change','No'],

  ['Sunny','Warm','High','Strong','Cool','Same','Yes']

]


columns = ['Sky','AirTemp','Humidity','Wind','Water','Forecast','EnjoySport']

df = pd.DataFrame(data, columns=columns)


print("Training Examples:\n")

print(df)
```

```python
# Initialize hypothesis
hypothesis = ['Ø'] * (len(columns)-1)

print("\nInitial Hypothesis:")
print(hypothesis)

# FIND-S Algorithm
for i, row in df.iterrows():
    if row['EnjoySport'] == 'Yes':
        example = row[:-1].tolist()

        if hypothesis[0] == 'Ø':
            hypothesis = example.copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != example[j]:
                    hypothesis[j] = '?'

        print(f"\nAfter Positive Example {i+1}:")
        print(hypothesis)

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)
```

```
Training Examples:

    Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool     Same        Yes

Initial Hypothesis:
['Ø', 'Ø', 'Ø', 'Ø', 'Ø', 'Ø']

After Positive Example 1:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

After Positive Example 2:
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

After Positive Example 4:
['Sunny', 'Warm', '?', 'Strong', '?', 'Same']

Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', 'Same']
```

Result

The FIND-S algorithm was successfully implemented in Google Colab. The final hypothesis obtained is